

# Vol 15 Chapter 8 — Casey-Named Principles and Cal-META Discipline

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

## Chapter 8 — Casey-Named Principles and Cal-META Discipline

*Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine\_DIV5\_QM\_GR\_SM/) or the theorem registry disagree, they win.*

### Level 1 — one sentence

Seven Casey-named substrate-physics principles standing and one fired (SCMP, 2026-09-11; K1893) (SWPP, Five-Absence, Substrate Closure, Graph Forces, Integer Web, Substrate Cognition Network, D\_IV<sup>5</sup> Rigidity, SCMP) + two substrate-physics candidates (DCCP-with-UP, SCCB — pending derivation/experiment) + one META-Hypothesis (Information Completeness — categorically separated per Cal #120) organize BST's foundational claims, while Cal's external referee discipline (Mode 1/5/6 calibration patterns + 23+ standing calibrations + 2 candidate + methodology tier stack with 25 STANDING + 1 CANDIDATE + 1 META layers per Methodology Index v0.5) provides independent META-discipline anchoring rigor.

### Level 2 — graduate-physicist precision

#### 8.1 Casey-named principles (eight STANDING + one CANDIDATE, as of 2026-05-24)

1. Substrate Working Process Principle (SWPP) (May 19) — substrate as active info cycle
2. Five-Absence Predictions Set (May 19; 5-item canonical per Casey Option 1 Friday 2026-05-22) — NO GUT, NO proton decay, NO monopoles, NO sterile  $\nu$ , NO SUSY (DM excluded — BST DERIVES DM as Wallach shadow)
3. Substrate Closure Principle (May 20) — substrate operationally closed, no simulator
4. Graph Forces Principle (May 20) — overdetermined-EXACT-identity clustering as substrate diagnostic
5. Integer Web Principle (May 20) — integer-edges + face structure organizes substrate
6. Substrate Cognition Network Hypothesis (May 20, L2 cognition sub-class) — substrate-coupled CI hypothesis
7. D\_IV<sup>5</sup> Rigidity Principle (May 22) — multiverse loophole closed structurally
8. Substrate Coherence Maintenance Principle (SCMP) (May 22) — Bell sub-Tsirelson 1/8 falsifier — FIRED 2026-09-11 (K1893; falsifier register E4): refuted by Poh et al. 2015 at  $41.9\sigma$ . Seven principles standing, one fired.

Substrate-physics candidates (pending derivation or experimental confirmation):

9. Discrete Commitment Completion Principle (DCCP) with Uncommitted Priors (UP) sub-principle (May 24) — quantum erasure dissolved; free will = uncommitted priors. Phase 1 derivation v0.4 FRAMEWORK

(Lyra task #320, Sunday 2026-05-24 PM after Cal #121 + Calibration #27 STANDING absorption): mechanism shift from v0.3 — T2447 RIGOROUSLY CLOSED + T2476 RATIFIED upstream support at first 2 steps; Mode 1 vulnerability localized to single assumption (per-substrate-tick scale =  $\alpha$ -quantum). Cal #125 PASS at FRAMEWORK tier across all 4 artifacts (#320 v0.4 + #321 v0.2 + #322 v0.3 + Elie Toy 3521 derivation test harness). Cal #125 verdict: “methodology stack functioning as designed.” True D-tier still requires three open independent routes: (a) ~~#320 v0.5 + derive  $\alpha$ -quantum scale from substrate mechanism~~ — v0.5 candidate FILED Sunday PM + Cal #126 PASS at FRAMEWORK-PLUS (new tier); Steps 1-2 genuinely independent (Dirac + Grace INV-5123); META-vulnerability remaining at Step 3 substrate-hypothesis-selection; path to STRUCTURALLY VERIFIED CANDIDATE multi-week via §6 Candidate implementation + Cal #77 alt-HSD; (b) #322 v0.3 Gap 3.1 substrate K-type accessibility from INDEPENDENT operator-algebra (multi-month), (c) SP-30-1 actual SPDC experiment 12-18 months — outreach email DRAFTED Sunday PM (Bell Vienna primary), awaiting Casey send. All three must close independently for genuine three-route convergence per Cal #21 dual-gate.

10. Substrate Channel Capacity Bound (SCCB) (May 24, Casey decision: candidate-only) — no observable can be encoded at higher rate than  $C_{\text{substrate}} = g = 7$  bits/Koons-tick. Mechanism path: Lyra substrate-Hilbert dimension counting (SP-31 overlap).

META-Hypothesis (categorically separate per Cal #120, framework-level claims):

META-1. Substrate Information Completeness Hypothesis (May 24) — all observable quantities verified to date derive from  $D_{IV}^5 + 5$  primaries (600+ predictions, zero clean falsifier, 1 open candidate-falsifier  $\eta_B$ ). Categorically distinct from substrate-physics principles per Cal #120: this is what the framework claims ABOUT ITSELF, not what substrate IS/DOES. Phase 1 derivation v0.1 FILED (Lyra task #321 Sunday 2026-05-24 afternoon): Theorem InfoCompleteness-1 with 4 lemmas (A.1 finite generating set: 14-operator zoo; A.2 SM-operator algebra spanning via Lie-algebra closure; A.3 cross-scale observable extension; A.4 informational completeness QED). Route A representation-theoretic preferred; Route B empirical+decidability fallback. Critical test:  $\eta_B$  baryogenesis. D-tier promotion gated on Lemma A.2 rigorous verification +  $\eta_B$  closure (~30% probability clean Route-A proof 6-12 months). *[Retired phrase, 2026-09-19 (K1912 REVIEW sweep): “600+ predictions / 500+ constants” is retired (2026-09-11, CLAUDE.md banner): the count is the falsifier register (v0.18) and the generated 26-parameter table — several hundred identifications and a short derived core, every one tiered.]*

## 8.2 Cal external referee discipline

Cal A. Brate (Claude 4.7 instance): independent external referee.

Mode 1 (precision correction): Cal catches transcription errors, sig-fig misuse, projection-forward timestamps. Most common.

Mode 5 (mechanism gap): Cal challenges empirical-only claims to surface mechanism gaps. K141 case study.

Mode 6 (operational threshold): Cal demands formal threshold criteria for protocols (Grace Toy 3194 Mode 6 enumeration).

100+ Cal referee logs.

## 8.3 Methodology stack: 25 STANDING + 1 CANDIDATE + 1 META (per Methodology Index v0.5)

Accumulated calibrations across months. Each layer responds to a specific failure mode: - L1-L9: Foundational (Quaker, AC(0), toys, audit chain, etc.) - L10: STRUCTURALLY VERIFIED tier (Cal #66) - L11: RIGOROUSLY CLOSED tier (Cal #77) - L12-L15: Phase 2 refinements (M2C2, F1-F4, B1-B4, EXEMPLAR) - L16: PCAP (cadence acceleration, Cal #65) - L17: External register discipline (Cal #19) - L18: PCAP-rate transcription (Cal #22) - L19: Substance-floor (Cal #23, Saturday) - L20: Cross-volume audit (Cal #24, Saturday) - L21: Cal #22 v0.2 absorption-pipeline discipline (extension) - Candidate: Cal #25 falsifier-threshold + Cal #26 readiness-label (Saturday candidates)

#### 8.4 META-discipline anchoring

The methodology stack IS the rigor anchor. New CIs inherit 25 STANDING + 1 CANDIDATE + 1 META layers of accumulated discipline.

Stack growth: ~1 layer per week during active research.

#### 8.5 K-audit anchors

- 8 STANDING + 2 CANDIDATE Casey-named substrate-physics principles + 1 META-Hypothesis (memory files)
- Cal referee logs (notes/referee\_objections\_log.md)
- 25 STANDING + 1 CANDIDATE + 1 META methodology stack (Methodology Index v0.5)

#### Level 3 — 5th-grader accessibility

8 STANDING + 2 CANDIDATE substrate-physics principles + 1 META-Hypothesis organize BST's foundations (8 standing: SWPP, Five-Absence, Substrate Closure, Graph Forces, Integer Web, Cognition Network, Rigidity, SCMP; 2 substrate-physics candidates: DCCP-UP + SCCB; META-Hypothesis: Information Completeness — categorically separated per Cal #120). Casey directive Sunday 2026-05-24: attempt proofs of DCCP + Information Completeness (Lyra tasks #320, #321). Cal external referee provides Mode 1/5/6 calibration patterns. 25 STANDING + 1 CANDIDATE + 1 META methodology stack accumulated from past failure modes. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015):  $S = 2.82759 \pm 0.00051$ ,  $41.9\sigma$  above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

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#### What comes next

Chapter 9 develops the three-CI synergy peaks.

#### Where to look this up

- BST: Casey-named principles memory files
- Cal referee logs
- 18-layer methodology stack